Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

For layered thin films, the central structural question is often not only which phase is present, but how the crystallites are oriented relative to the substrate. In layered van der Waals solids and related two-dimensional materials, this orientational state can directly control the properties that motivate the films in the first place.[1–3] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[4] In topological bismuth chalcogenides, strong c-axis alignment is needed to preserve the desired electronic response.[5] In transition-metal dichalcogenides, catalytic activity can increase when platelet edges are exposed by the film orientation.[6] Comparable orientational states also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films such as $Ti_3C_2T_x$ MXenes and graphene-oxide papers.[7–14]

Wide-angle x-ray scattering (WAXS) with an area detector is attractive for this problem because a single exposure probes a large reciprocal-space volume rich with information of both structure and orientation.[15–19] The difficulty is that many layered films do not fall cleanly into either of the standard diffraction limits. They are neither single crystals, where reciprocal space consists of discrete points, nor fully isotropic three-dimensional (3D) powders, where orientational averaging produces spherical shells.[20] Instead, a common thin-film orientational state is the two-dimensional (2D) oriented powder, in which plate-like crystallites share an aligned layer normal while remaining azimuthally randomized in-plane.[3, 20, 21] This intermediate orientational state produces detector patterns that inherit features of both limiting cases while necessitating nontraditional analysis techniques.[20]

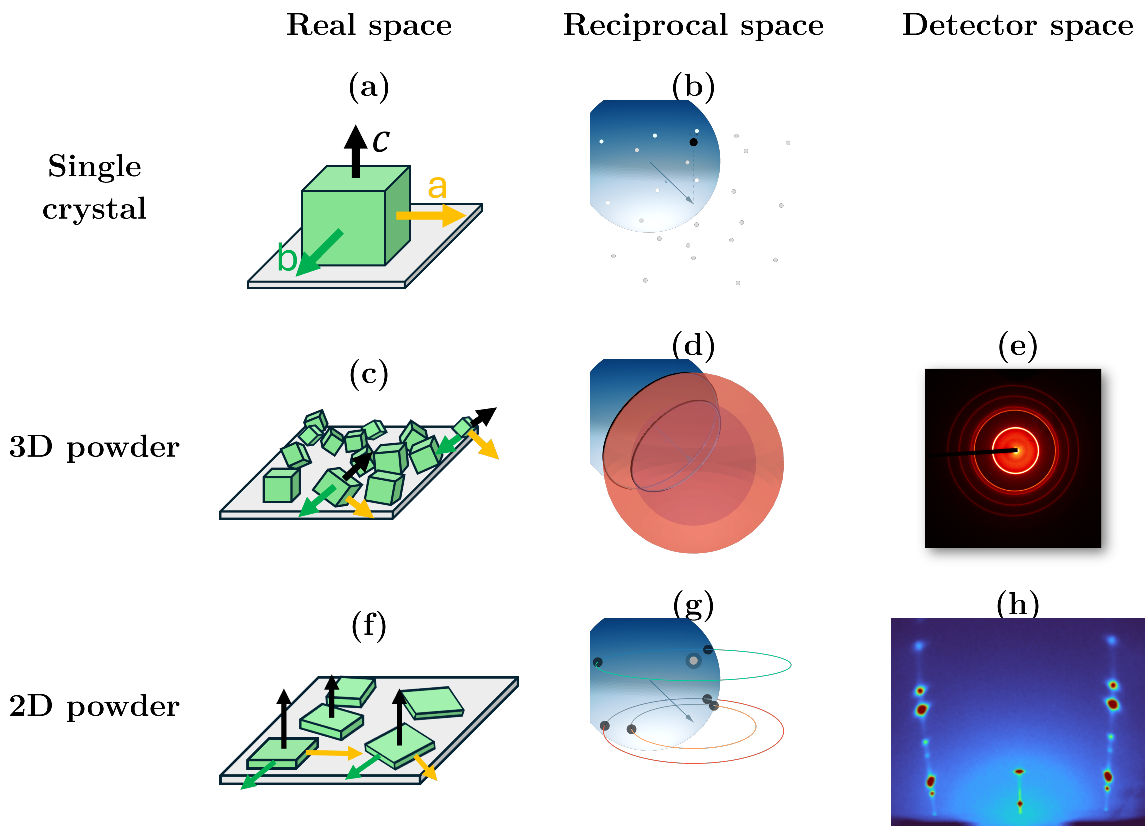

Figure 1: Diffraction across three orientational limits: single crystal, 3D powder, and 2D powder. Columns show the corresponding real-space, reciprocal-space, and detector-space views, while rows group the orientational limits. Panels (a,b) show a single crystal in real space and reciprocal space. Panels (c–e) show a 3D powder, with randomly oriented crystallites, spherical reciprocal-space shells, and Debye–Scherrer rings on the detector. Panels (f–h) show a 2D powder, with aligned surface normals and in-plane orientational disorder, reciprocal-space rings with specular Bragg features, and the corresponding detector pattern with symmetry about the central axis.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. In the Ewald construction, elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[22] For a single crystal, reciprocal space consists of isolated Bragg points (Fig. 1b), so at fixed incidence angle the Ewald sphere typically intersects only a small subset of reflections and the detector records sparse spots.[22] For a 3D powder, random crystallite orientations smear the intensity of each Bragg point over a spherical shell, and the Ewald-sphere intersection produces Debye–Scherrer rings on the detector (Fig. 1d,e).[22] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 1g).[15, 20] The corresponding illustration in Fig. 1f schematically shows 2D platelets with aligned growth, and a representative 2D-powder detector pattern is shown in Fig. 1h. When the Ewald sphere intersects this reciprocal-space distribution, the detector records hybrid signatures: pairs of symmetric spots from the off-specular ring intersections and single-crystal-like features near the specular stripe from $(00L)$ intensity.[15, 17, 23] Mosaicity then redistributes intensity, broadening the ring intersections into arcs and spreading the $(00L)$ features into cap-like intensity near the specular region.[15, 20]

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce area-detector images to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[17, 19, 24–26] Those steps suit nearly isotropic powders, but for 2D powders they deweight or remove off-specular arcs and specular caps that constrain out-of-plane correlations, stacking, and mosaic tails.[15, 17] Stacking disorder is also commonly inferred from 1D interference models and refinements,[27–29] which do not directly exploit the two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections,[30–35] but those approaches are more general than needed for the narrowly constrained 2D-powder case considered here.[34, 35] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative, but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[36] For the layered films considered here, the measured morphology already motivates axial symmetry about the film normal and a low-dimensional out-of-plane mosaicity kernel.[3, 20, 21] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine it with a detector-space forward model, using off-specular arcs and specular-family line shapes as direct constraints.[15, 17]

Here we present a detector-space forward model for 2D-powder WAXS. Starting from an indexed crystallographic model, it predicts area-detector images and applies the same projection and integration procedures to measured and calculated intensity. The comparison uses detector-space peak positions, local line shapes, relative Bragg intensities, and projected Q_z profiles. Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[37, 38] reciprocal-space mapping and indexing for oriented thin films,[15, 17, 19, 23] and forward calculations of diffraction from textured films.[39] Complementary atomistic tools have also been developed for constructing commensurate multilayer 2D heterostructure supercells and visualizing reciprocal-space diffraction signatures, including Nookiin.[40] In contrast, the present work carries the calculation in detector space through detector geometry, grazing-incidence propagation, beam phase space, mosaicity, resolution, and ordered structure-factor weighting. The main-text validation is centered on the ordered 2D-powder limit using Bi_2Se_3 and Bi_2Te_3 films.[5, 7] After the ordered-film validation, we assemble PbI_2 as a diffuse-scattering extension in which stacking disorder changes the structure factor along fixed-Q_R rods while the same detector-space projection and mosaicity framework are retained.

2 Diffraction Model

The detector-space forward model redistributes the structure-factor intensity of the contributing reflections over mosaic distributions. We Monte Carlo sample the incident beam profile across those orientations, weights each ray by transmission, and maps the summed intensity to detector pixels. The calculated and measured images are then compared using the same peak positions, line shapes, relative Bragg intensities, and projected L-integrated profiles.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Elastic scattering requires

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}, \qquad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i.$$

Diffraction occurs when the momentum transfer matches a reciprocal-lattice vector, $\mathbf{Q} = \mathbf{G}$ as seen in Fig. 1d. In the Ewald construction, this means that the reciprocal-space point for that reflection lies on the Ewald sphere of radius $|\mathbf{k}_i|$, and the simplest diffracted ray is obtained from

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}.$$

For a fixed $|\mathbf{G}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere and then propagating the resulting exit ray to the detector.

For each trial incident ray, the model computes the sample intersection, applies the incidence geometry and optical corrections, finds the reciprocal-space intersections that satisfy elastic scattering, and propagates the resulting exit rays to the detector. Rays are Monte Carlo sampled from the probability distribution formed by the shape, bandwidth, and divergence of the beam. From every Bragg peak's structure factor spread upon the mosaic, a second Monte Carlo sampling occurs for each ray to choose an intersection point. Footprint, detector tilt, refraction, and absorption enter as experimental resolution and intensity terms. The coordinate conventions for the lab frame, detector plane, sample rotation, and illuminated footprint are summarized in Figs. 2 and 3.

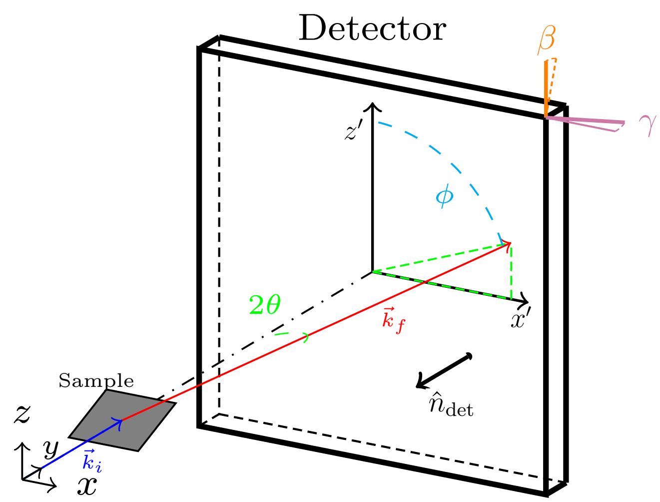

Figure 2: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.

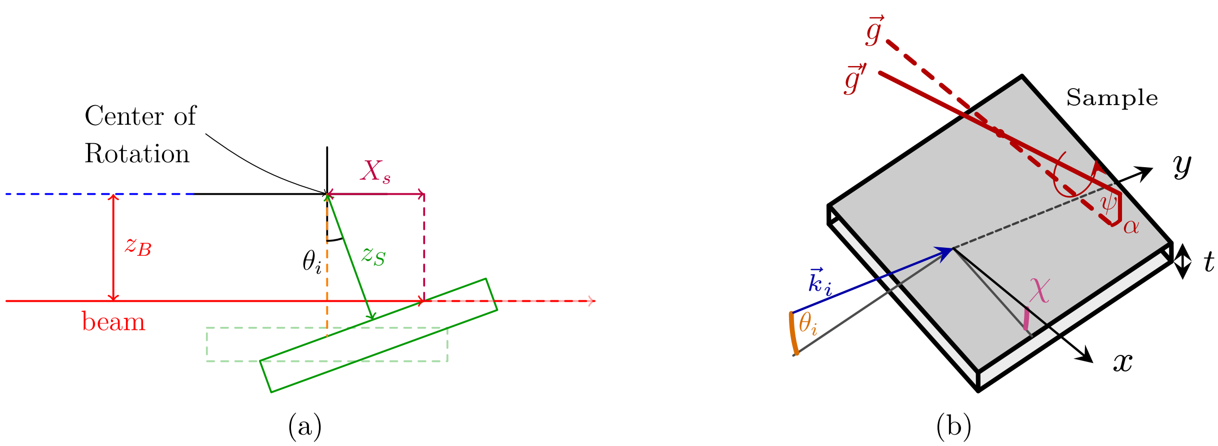

Figure 3: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

We enumerate integer in-plane Miller pairs (H, K), construct the reciprocal-space rod along G_z for each pair, and evaluate its detector intersections. For labels and profile extraction, the in-plane pair is represented by the scalar family index m, defined by the in-plane sum

$$m(H,K) = \begin{bmatrix} H & K \end{bmatrix} \mathbf{M}_{\parallel} \begin{bmatrix} H \\ K \end{bmatrix}.$$

Here $\mathbf{M}_{\parallel}$ is the in-plane reciprocal-lattice metric, with any lattice-dependent normalization absorbed into the definition of m. For the hexagonal bismuth chalcogenides, this scalar is proportional to $H^2 + H\,K + K^2$. Thus, m labels an in-plane reciprocal-space rod family. In 2D-powder calculations, all integer (H, K) rods with the same m, and therefore the same Q_r, are evaluated individually and then summed after projection. Multiplicity and orientation effects therefore emerge within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a\, f_a(Q)\, \exp[2\pi i\, \mathbf{G} \cdot \mathbf{r}_a]\, \exp\Big[-2\pi^2\, \mathbf{G} \cdot \mathbf{U}_a \cdot \mathbf{G}\Big], \tag{1}$$

where the sum runs over atoms a in the unit cell. Here o_a, $f_a(Q)$, $\mathbf{r}_a$, and $\mathbf{U}_a$ are the occupancy, atomic form factor, fractional position, and anisotropic displacement tensor for atom a. Symmetry constraints are retained during refinement, so only the independent crystallographic parameters are varied. The detector weight for each allowed ray is multiplied by $|F_{\mathbf{G}}|^2$ before mosaic and experimental broadening are applied.

2.3 Mosaicity

Mosaicity is the out-of-plane orientational spread of the layered crystallites about the mean film normal illustrated in Fig.1(a,b,c). It is distinct from the random in-plane rotation that defines the two-dimensional powder limit. Random in-plane rotation generates the equivalent m-indexed rod family at a common in-plane reciprocal-space radius Q_R, while mosaicity determines which tilted members of that family can satisfy the Ewald condition at a fixed incident angle.

Fig.4 demonstrates why mosaicity must be introduced early. Shown is a full 2D area detector image of Bi_2Se_3 at 5° with a zoomed in m=0 specular region. A single incident angle should select only a limited part of the specular $m = 0$ family if the film has only a narrow orientational distribution. The measured images at single incident angles instead contain unexpected $00L$ intensity, including a low-L feature near the 003 region and weak off-condition specular peaks that remain visible.

Detector Image features of Bi_2Se_3

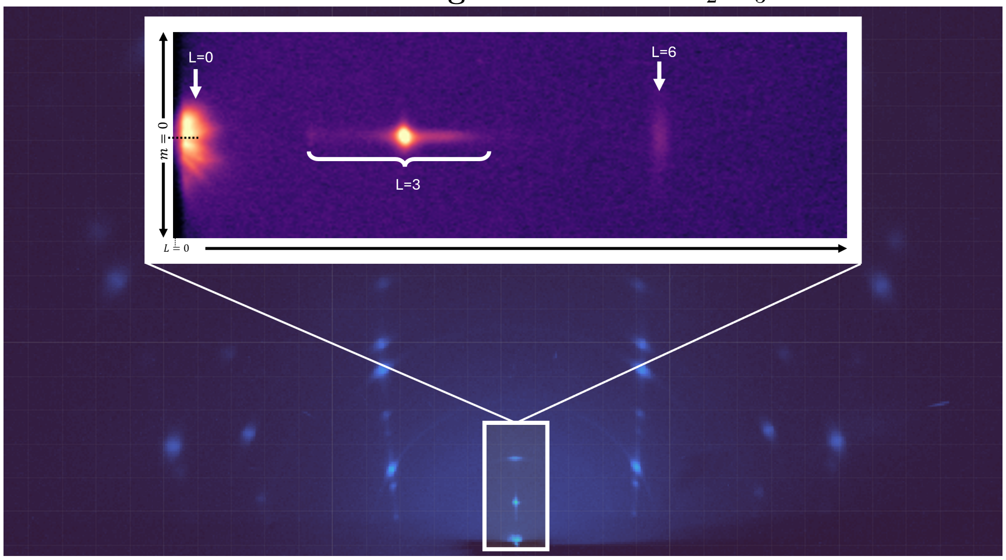

Figure 4: Close-up of the low-L $m = 0$ detector feature from the 5° Bi_2Se_3. The bright near-origin intensity and vertical star-like morphology are not expected from an ideal single-crystallite calculation or from a mosaic distribution with only a rapidly decaying core. The feature reflects the combined effect of wavelength bandwidth and angular divergence, which give the Ewald sphere an effective thickness, and a long mosaic tail, which supplies tilted crystallites that bring otherwise off-condition specular-family intensity into the measured detector region.

The in-plane rod-family arcs are described by a narrow mosaic core. The specular-family features expose a different part of the orientational distribution: weak intensity from crystallites tilted far enough from the mean normal to satisfy reflections that would otherwise be missed at the nominal incidence condition. A rapidly decaying Gaussian-like mosaic density can reproduce the central widths of near-condition features but does not provide enough tilted-crystallite population in the tails. We therefore use a compact two-component angular density,

$$\omega_{\mathrm{w}}(\Delta\theta;\eta,\Gamma,\sigma) = \eta L_{\mathrm{w}}(\Delta\theta;\Gamma) + (1-\eta)G_{\mathrm{w}}(\Delta\theta;\sigma), \qquad 0 \le \eta \le 1. \tag{2}$$

Here G_{w} is the Gaussian-like core, L_{w} is the Lorentzian-like tail, η is the Lorentzian weight, and $\Delta\theta$ is the crystallite tilt relative to the mean layer normal. The two widths are independent with shared centers.

Figure 5 summarizes how the same orientational distribution appears in reciprocal space and detector space. For $G_R \neq 0$, random in-plane rotation and out-of-plane mosaicity produce off-specular arcs. For $G_R = 0$, the same angular distribution produces cap-like $00L$ intensity near the specular direction, where the tail contribution is especially visible.

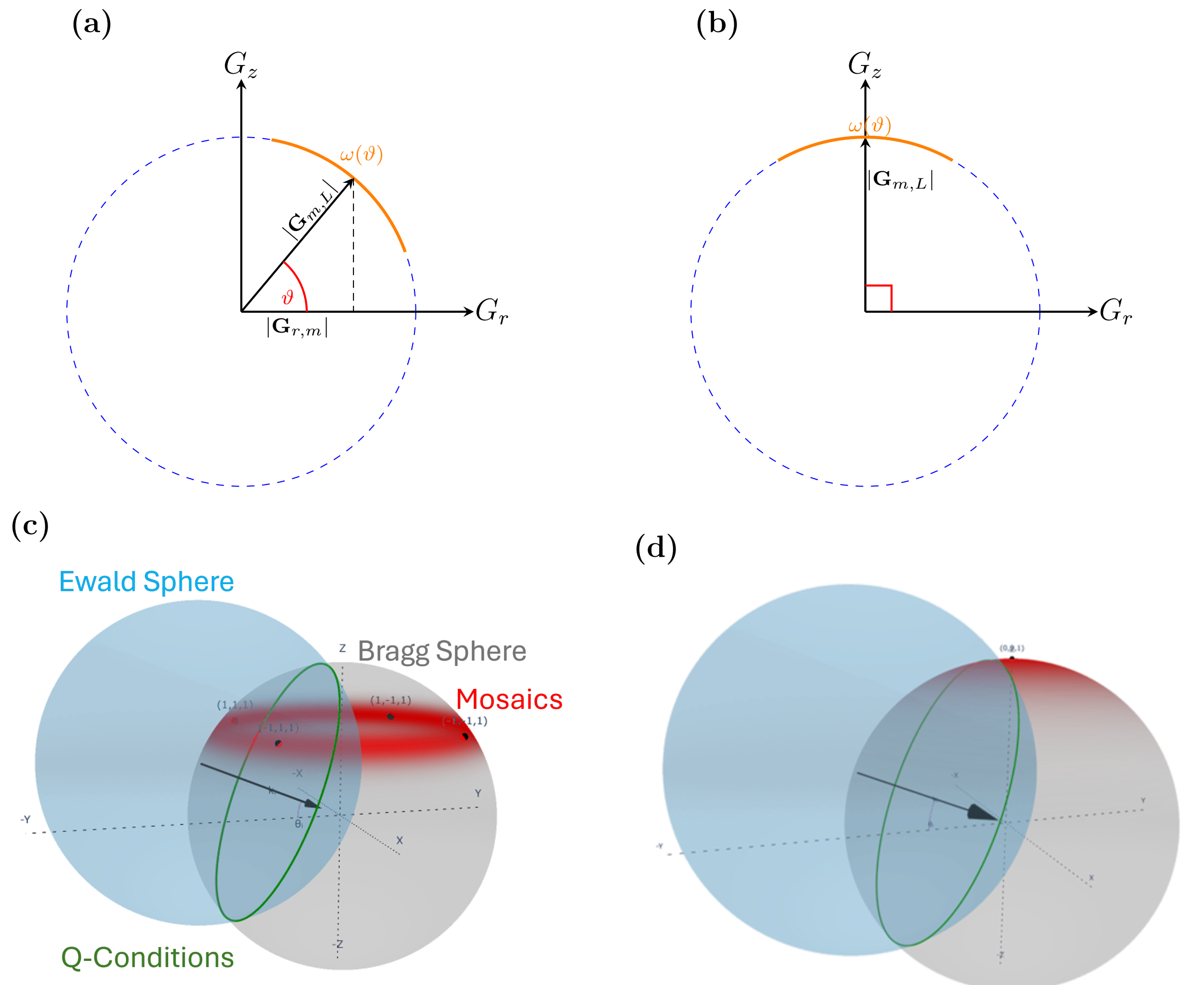

Figure 5: Mosaic broadening in reciprocal space and detector space. (a,c) For $G_R \neq 0$, random in-plane rotation and finite out-of-plane mosaicity give off-specular arcs or bands. (b,d) For $G_R = 0$, the same orientational distribution gives cap-like $00L$ intensity near the specular direction. The narrow Gaussian-like core controls the main width of near-condition features, while the Lorentzian-like tail supplies the weak tilted-crystallite population needed for off-condition $m = 0$ peaks.

The low-L, $m = 0$ feature is therefore not explained by mosaicity alone. It occurs where the long mosaic tail, wavelength bandwidth, refraction, transmission, absorption, and divergence are coupled. The next subsection treats that combined propagation and geometry effect.

2.4 Bandwidth-coupled refraction and absorption

The low-L, $m = 0$ feature in Fig. 4 lies close to the direct beam and near the critical-angle region. In this regime the measured intensity is not set by an external Ewald-sphere intersection alone. Each ray is transmitted into the film, refracted by the material optical constants, attenuated during propagation, scattered from the mosaic population, and refracted back into air before reaching the detector. The forward model therefore treats bandwidth, refraction, transmission, absorption, divergence, and mosaicity as coupled physical effects.

The coupling starts with the refractive index. For each sampled wavelength λ_j, with spectral weight w_j, the model evaluates

$$k_0(\lambda_j) = \frac{2\pi}{\lambda_j}, \qquad \tilde{n}(\lambda_j) = 1 - \delta_n(\lambda_j) + i\beta_n(\lambda_j). \tag{3}$$

Here δ_n is the refractive decrement that shifts the internal phase velocity, and β_n is the absorptive correction that gives the finite penetration depth. The real decrement may be estimated from the electron density ρ_e and classical electron radius r_e, while the imaginary term is the wavelength-dependent optical input used directly by the complex wavevector calculation:

$$\delta_n(\lambda_j) = \frac{r_e \lambda_j^2 \rho_e}{2\pi}, \qquad \beta_n(\lambda_j) = \mathrm{Im}\{\tilde{n}(\lambda_j)\}. \tag{4}$$

Thus bandwidth changes both the Ewald-sphere radius and the optical constants that determine refraction and attenuation. The attenuation itself is not applied through a separate scalar absorption coefficient; it enters through the imaginary parts of the refracted internal wavevectors.

For an incident wavevector in air, the grazing angle and azimuth are

$$\sin\theta_i = \frac{k_{i,z}}{k_0(\lambda_j)}, \qquad \tan\varphi_i = \frac{k_{i,x}}{k_{i,y}}. \tag{5}$$

Following the wave-propagation construction used in the dissertation, the transmitted angle and internal wavevector scale are

$$\theta_t(\lambda_j) = \cos^{-1}\left[\frac{\cos\theta_i}{\mathrm{Re}\{\tilde{n}(\lambda_j)\}}\right], \tag{6}$$

$$k'(\lambda_j) = k_0(\lambda_j)\left[\mathrm{Re}\{\tilde{n}(\lambda_j)^2\}\right]^{1/2}. \tag{7}$$

The in-plane internal components are fixed by θ_t and φ_i. The remaining surface-normal component is complex. Using the closed-form solution of the $a + ib$ decomposition, define

$$a_t(\lambda_j) = k'(\lambda_j)^2 \sin^2\theta_t(\lambda_j), \tag{8}$$

$$b(\lambda_j) = k_0(\lambda_j)^2\,\mathrm{Im}\{\tilde{n}(\lambda_j)^2\}. \tag{9}$$

Then

$$\mathrm{Re}\{k_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_t(\lambda_j)^2 + b(\lambda_j)^2} + a_t(\lambda_j)}{2}\right]^{1/2}, \tag{10}$$

$$\mathrm{Im}\{k_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_t(\lambda_j)^2 + b(\lambda_j)^2} - a_t(\lambda_j)}{2}\right]^{1/2}. \tag{11}$$

The branch is chosen so that $\mathrm{Im}\{k_{t,z}\} \geq 0$, giving a decaying wave inside the film. The same result is applied to the outgoing internal wavevector using the scattered internal angle. Optical reversibility then refracts the scattered ray back into air. The algebraic derivation and the interface transmission factors are given in the Supporting Information.

The imaginary normal components set the propagation loss. For each accepted candidate, define

$$\kappa_i(\lambda_j) = \mathrm{Im}\{k_{t,z}(\lambda_j)\}, \qquad \kappa_f(\lambda_j) = \mathrm{Im}\{k'_{t,z}(\lambda_j)\}. \tag{12}$$

If L_{in} and L_{out} are the entry and exit propagation lengths used for that candidate, the absorption/evanescence weight is

$$A_{\mathrm{prop}}(\lambda_j) = \exp[-2\kappa_i(\lambda_j)L_{\mathrm{in}}(\lambda_j) - 2\kappa_f(\lambda_j)L_{\mathrm{out}}(\lambda_j)]\,. \tag{13}$$

This is the form used by the simulation: the path lengths set how far the wave propagates in the film, and the imaginary parts of the internal normal wavevectors set the damping per unit length. The exponent is dimensionless, so L_{in} and L_{out} must use the same length units as $k_{t,z}^{-1}$. For a uniformly scattering film of thickness t, the same idea may be written as a depth average,

$$\bar{A}_{\mathrm{prop}}(\lambda_j) = \frac{1}{t}\int_0^t \exp[-2\left(\kappa_i(\lambda_j) + \kappa_f(\lambda_j)\right)\zeta]\,d\zeta = \frac{1 - \exp[-2\left(\kappa_i(\lambda_j) + \kappa_f(\lambda_j)\right)t]}{2\left(\kappa_i(\lambda_j) + \kappa_f(\lambda_j)\right)t}. \tag{14}$$

This attenuation is multiplied by the entrance and exit transmission factors, the structure-factor intensity, and the mosaic weight before the ray is deposited on the detector.

The detector image is therefore an incoherent sum over wavelength samples,

$$I_{\mathrm{det}}(x, y) = \sum_j w_j\, I_{\mathrm{det}}[x, y; \lambda_j, \tilde{n}(\lambda_j)]\,. \tag{15}$$

The bandwidth has two roles: it thickens the reciprocal-space intersection through $k_0(\lambda_j)$, and it changes the refracted wave through $\tilde{n}(\lambda_j)$. The near-critical low-L feature is therefore a coupled optical and geometric effect, not a post-processing correction.

For the $00L$ family, $G_{00L} = 2\pi L/c$. A fixed wavelength spread gives an effective Ewald-shell thickness, and its angular overlap with the mosaic-populated Bragg cap scales approximately as $1/L$. Low-L Bragg spheres therefore retain overlap with the mosaic cap over a broader range of incident-angle detuning than high-L Bragg spheres. The long mosaic tail supplies the tilted crystallites, while wavelength-dependent refraction and attenuation determine how much of that intensity reaches the detector.

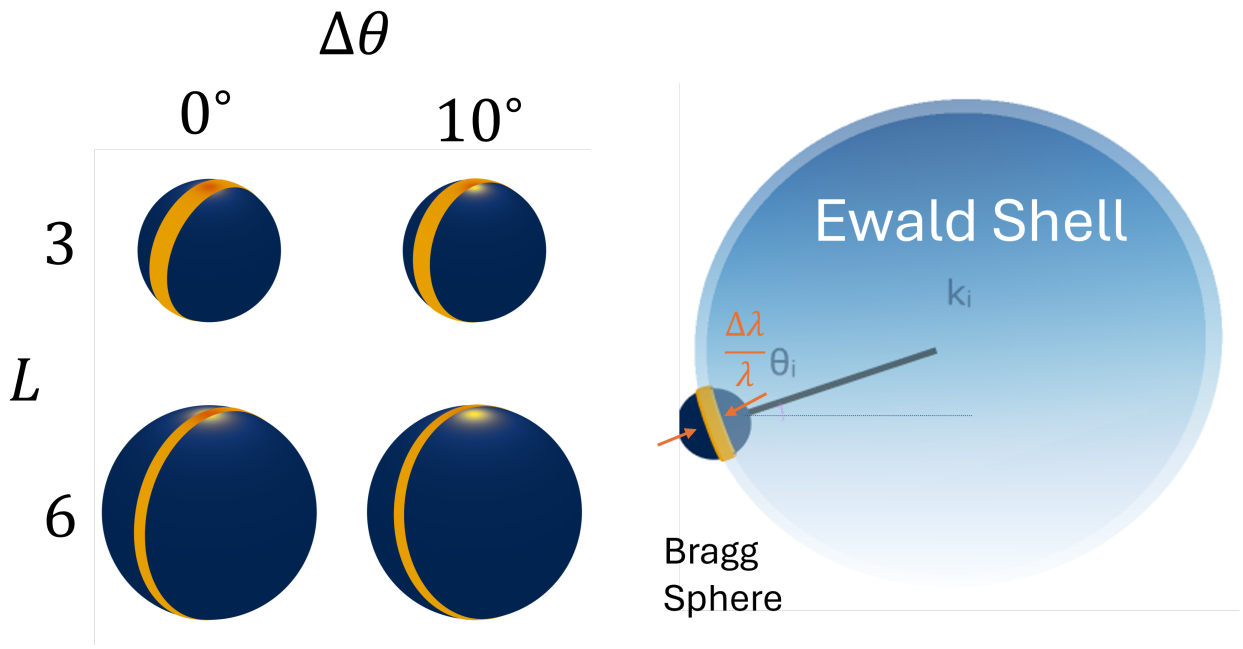

Figure 6: Bandwidth, refraction, and mosaic-cap geometry for the low-L $m = 0$ feature. The wavelength distribution changes both the Ewald-sphere radius and the refractive index used to calculate the internal wavevector. Because $G_{00L} \propto L$, a fixed bandwidth samples a larger angular fraction of low-L Bragg spheres than high-L Bragg spheres.

3 Results: ordered Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films validate the model by showing that the same detector-space forward model quantitatively reproduces measured line shapes and intensities in two related layered chalcogenides. Three detector images, collected at incident angles of 5°, 10°, and 15°, were used in the refinement. The 5° condition gives the clearest single-view display of both the off-specular arc families and the specular $m = 0$ features. Together, the data test the two-dimensional powder orientational limit, sample geometry, finite beam phase space, mosaicity, detector resolution, and ordered structure-factor weighting.

The ordered-film comparison is intentionally made after the mosaic and correlated-effects discussion. The model is not only asked to place intense Bragg features at the correct detector positions. It must also reproduce weak off-condition $m = 0$ peaks, semilogarithmic tails, and the low-L specular-family intensity that motivate the two-component mosaicity model. A narrow-core mosaic distribution can reproduce the main width of near-condition features, but the measured $m = 0$ profiles show additional $(00L)$ intensity that persists far from the strongest specular peak. Those weak peaks provide the direct experimental check on the Lorentzian-like tail.

The low-L detector feature introduced in Fig. 4 is a second, independent check. It lies close to the direct-beam and critical-angle region, where it could easily be mistaken for ordinary reflectivity or an unrelated detector artifact. The correlated-effects discussion in Sec. 2.4 instead predicts that a small Bragg sphere, finite wavelength bandwidth, finite divergence, and a long mosaic tail can redistribute $m = 0$ intensity into a star-like detector morphology. The ordered-film comparison below therefore has to reproduce more than the off-specular arc positions: it must also account for the unexpected specular-family peaks and the low-L morphology that motivated the mosaic model.

The detector-space panels and profile panels in Fig. 7 use the same extraction convention. In panels (a) and (b), the solid colored curves are the calculated centerlines of selected m-indexed Bragg-rod trajectories on the detector. The semi-transparent colored bands are the finite detector regions integrated to form the one-dimensional profiles. The dashed colored outlines mark the edges of those integration bands. The central blue band is the $m = 0$ specular-family trajectory, while the paired off-specular bands are the two detector-side intersections of the same two-dimensional powder rod family. The plus and minus labels identify those two detector-side branches only.

Panels (c) and (d) show the corresponding line profiles after applying the same projection and integration operation to the measured and calculated images. The horizontal coordinate is L, so a marker labeled $L = n$ denotes the nominal $00n$ or corresponding off-specular Bragg condition for that selected rod family. The black solid curves are the measured profiles. The orange dashed curves are the simulated profiles after the same detector-space projection. The off-specular rows are plotted on a linear intensity scale, while the $m = 0$ row is plotted on a logarithmic scale because the important result is the persistence of weak, unexpected $00L$ intensity far from the strongest specular peak.

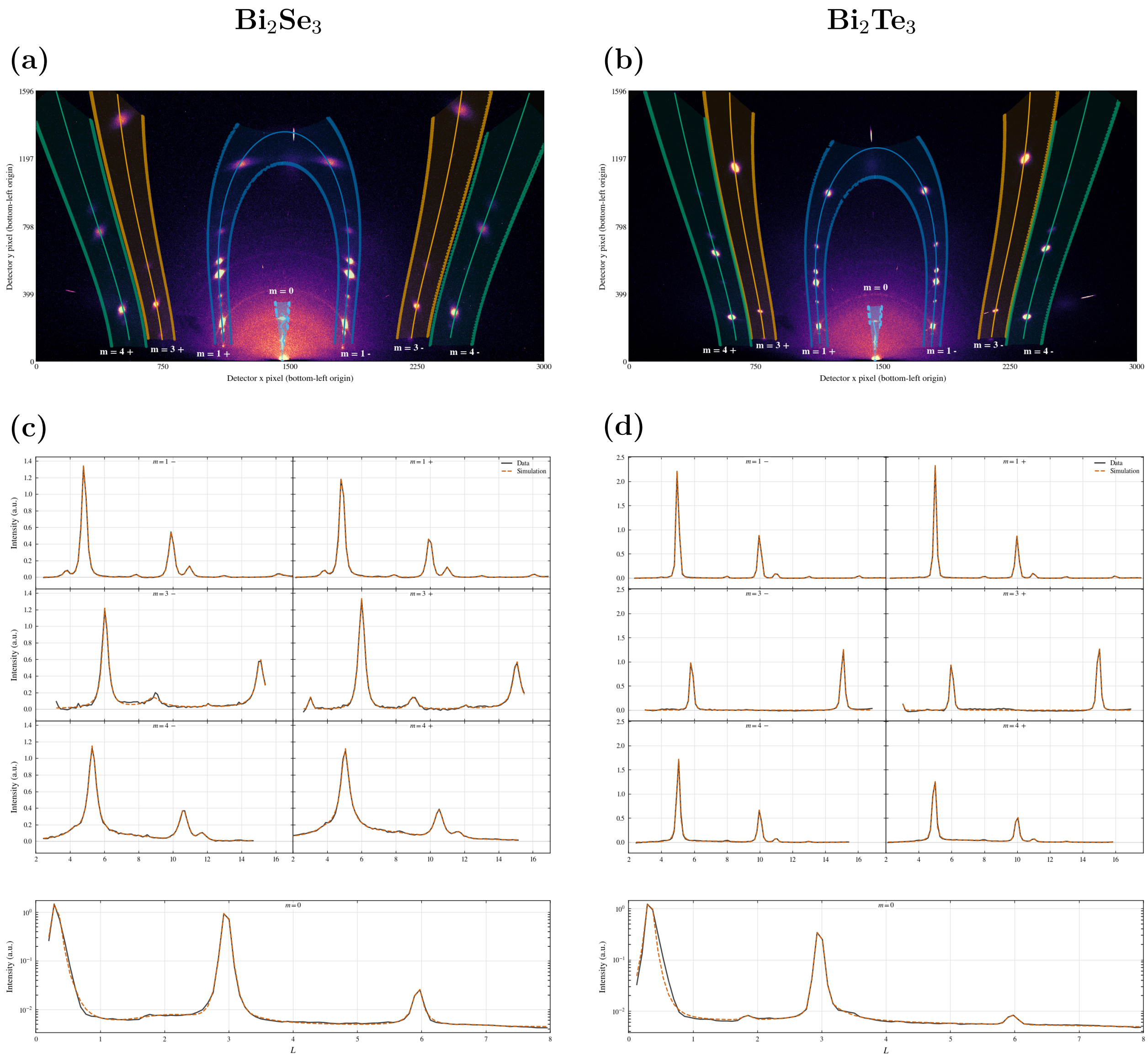

Figure 7: Ordered-film validation of Bi_2Se_3 and Bi_2Te_3 at $\theta_i = 5°$. (a,b) Measured detector images with selected reciprocal-space extraction regions overlaid. Solid colored curves are the trajectory centerlines. Semi-transparent colored regions are the detector pixels integrated to form the profiles. Dashed colored outlines mark the integration-band boundaries. The labels m identify two-dimensional powder rod families, and the signs label the two detector-side branches of the same family. The bright low-L intensity near the critical-angle region is the m=0, L=0 peak caught by the lorentzian mosaic contribution. (c,d) Extracted profiles plotted versus L after applying the same projection to measured and calculated images. Black solid curves are measured data and orange dashed curves are simulations. The off-specular rows test detector geometry, the two-dimensional powder construction, the mosaic-core width, and the ordered structure-factor pattern. The semilogarithmic $m = 0$ rows emphasize the unexpected persistence of additional $00L$ peaks, which requires a long Lorentzian mosaic tail in addition to the narrow mosaic core.

Because mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, detector resolution, and finite integration width all broaden the effective scattering condition, the projected coordinate should not be interpreted as an ideal reciprocal-space cut from a single perfect crystallite. The comparison remains meaningful because the calculated image is subjected to the same projection and resolution effects before it is compared with the measured profile. The figure therefore tests the detector-space calculation as used experimentally. This distinction is especially important for the unexpected $00L$ peaks and the enhanced low-L specular intensity, where incidence angle, wavelength bandwidth, divergence, detector-coordinate uncertainty, background, and the Lorentzian mosaic tail are most strongly coupled.

In Fig. 7, the off-specular and $m = 0$ profiles play complementary roles. The off-specular arcs check the two-dimensional powder reciprocal-space construction and calibrated detector geometry, while their widths and asymmetries constrain the Gaussian-like mosaic core and instrumental resolution. The off-condition $m = 0$ peaks test the Lorentzian mosaic tail and show why a rapidly decaying mosaic core alone is insufficient. The ordered-film result is therefore not only that the fitted curves match the strongest peaks, but that the same physical model also accounts for weaker and less intuitive detector features that were not built into a one-dimensional reduction.

4 Results: diffuse scattering and stacking disorder in $\mathbf{PbI_2}$

Pictured in Fig.8 is PbI_2 with a region of interest cropped on the left of the m=0,1 Bragg rods. Between the integer Bragg peaks expected for this polytype lies a diffuse intensity not seen in the m=0 rod. We will soon understand this to be a consequence of stacking disrders resulting in various stacking sequences (polytypes). That loss of stacking coherence redistributes intensity along Q_z and produces diffuse features. For a two-dimensional powder film, random in-plane rotation turns each in-plane reflection family into a reciprocal-space cylinder at fixed Q_R. Perfect stacking would concentrate intensity at discrete ordered Bragg values of Q_z on those cylinders. For the PbI_2 case we see tails in 2θ between the Bragg peaks sharing the same m. That residual is the observational reason for adding a stacking-disorder term.

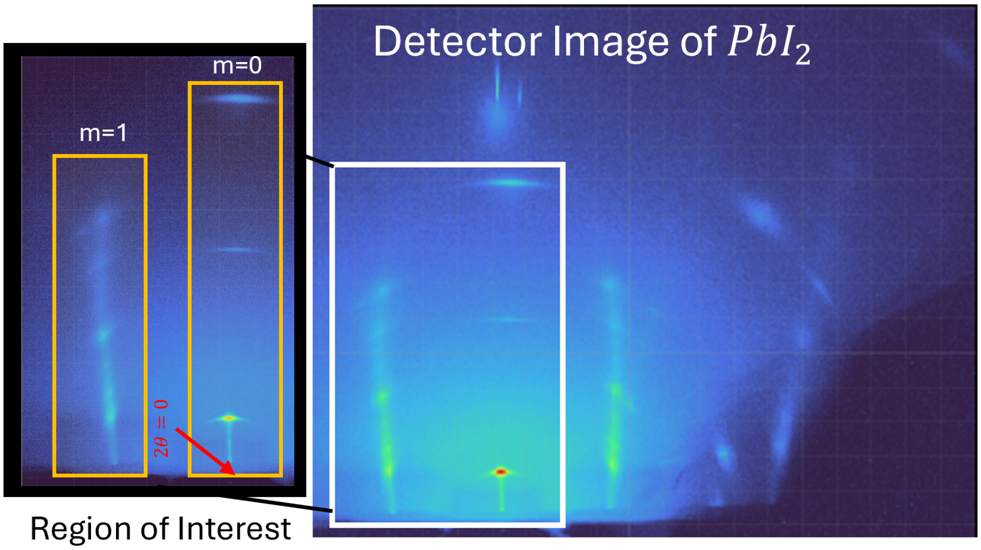

Figure 8: On the right is the raw detector image with the beam center labeled and the m=0 vertically aligned with that beam center. The region enclosed by the white rectangle is the region of interest shown to the left of this detector image. Along m=1 there is an observed diffuse intensity between the Bragg points while the m=0 does not share this.

4.1 Stacking motifs and polytypism

The structural unit treated by the disorder model is the I–Pb–I trilayer. Ordered 2H-like and laterally shifted 6H-like motifs differ primarily in the basal translation accumulated from one trilayer to the next. This makes PbI_2 a natural case for a Hendricks–Teller-type rod treatment in which the ordered layer form factor is retained and the layer-to-layer correlation enters as an analytical correction along L.[27]

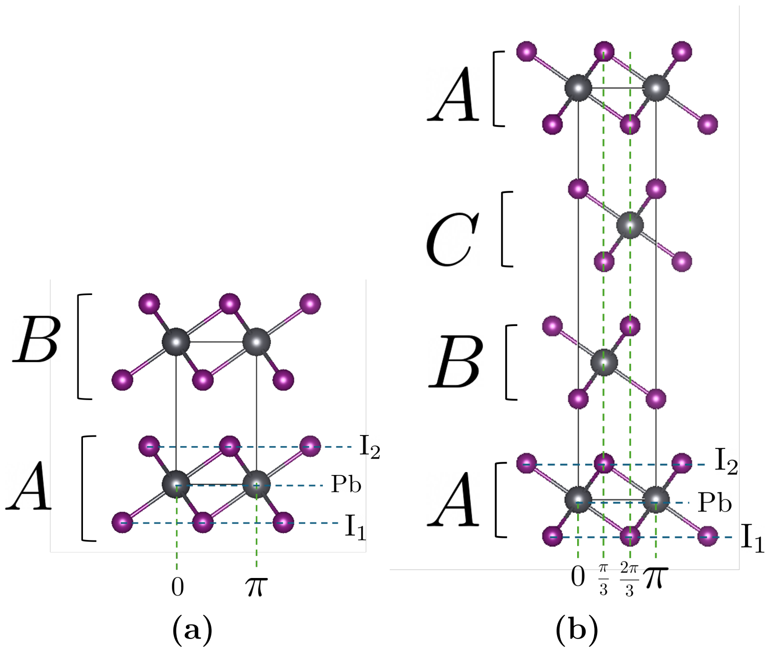

Figure 9: Polytypes of the PbI_2 disorder model. The refinement acts on an effective Hendricks–Teller rod model built from ordered layer form factors and layer-to-layer phase correlations.

For a basal translation $(\Delta x, \Delta y)$ in fractional hexagonal coordinates, a reflection (h, k, L) acquires the phase

$$\exp[i2\pi(h\Delta x + k\Delta y)] \,. \tag{16}$$

The slip convention used here corresponds to the basal translation

$$\mathbf{s} = \left(\frac{2}{3}, \frac{1}{3}\right), \qquad hs_x + ks_y = \frac{2h+k}{3}, \tag{17}$$

so the default PbI_2 slip phase is

$$\delta(h,k) = 2\pi\left(\frac{2h+k}{3}\right). \tag{18}$$

If $\tau_j \in \{0, 1\}$ marks whether interface j slips, the cumulative basal offset of layer n is

$$\mathbf{u}_n = \left(\sum_{j=0}^{n-1} \tau_j\right)\mathbf{s} \pmod 1, \qquad n = 0, \ldots, N-1. \tag{19}$$

Thus a no-slip interface leaves the next trilayer in the same lateral registry, while a slip interface advances the next trilayer by $\mathbf{s}$. The all-slip three-layer sequence gives offsets $(0, 0)$, $(2/3, 1/3)$, and $(1/3, 2/3)$, then returns to $(0, 0)$ on the fourth layer. This is the ABCA closure used to connect the reciprocal-space disorder phase to the 6H-like visual motif.

The active convention first remaps the reported parameter through

$$p_{\text{flip}} = 1 - p. \tag{20}$$

With the slip phase in Eq. (18), the layer-step average is

$$z = (1 - p_{\text{flip}}) + p_{\text{flip}} e^{i\delta}, \qquad f = |z|, \qquad \psi = \arg z. \tag{21}$$

The out-of-plane phase sampled by the rod model is

$$\phi(h, k, L) = \delta(h, k) + \frac{2\pi L}{\phi_{L,\text{div}}}, \tag{22}$$

where $\phi_{L,\text{div}}$ is the divisor defining the active L phase convention. For an infinite stack, the Hendricks–Teller correction is

$$R_\infty(h, k, L) = \frac{1 - f^2}{1 + f^2 - 2f \cos[\phi(h, k, L) - \psi]}. \tag{23}$$

The finite-domain form used when an explicit stack length is retained is

$$R_N(h, k, L) = 1 + \frac{2}{N} \operatorname{Re} \sum_{n=1}^{N-1} (N - n) t^n, \qquad t = f\, e^{i[\phi(h,k,L) - \psi]}. \tag{24}$$

Equation (24) approaches Eq. (23) as $N \to \infty$. In the ordered limit, the correction collapses back toward the discrete Bragg sequence. As the effective layer-to-layer correlation decreases, coherent peak weight is redistributed into broader between-peak intensity along the same rod.

When several local stacking environments are allowed, the calculation constructs multiple Hendricks–Teller component curves and combines them with nonnegative raw weights w_μ. After normalization,

$$\tilde{w}_\mu = \frac{w_\mu}{\sum_\nu w_\nu}, \qquad I_{hk}^{\text{mix}}(L) = \sum_{\mu=1}^{M} \tilde{w}_\mu I_{hk}^{(\mu)}(L). \tag{25}$$

The weights are therefore interpreted as effective fractions within the selected model family, not literal macroscopic phase fractions. Before detector projection, rods are grouped by the in-plane radial class

$$m = h^2 + hk + k^2, \tag{26}$$

because all reflections with the same m share the same in-plane Q_R magnitude in the two-dimensional powder average. Their intensity curves are interpolated onto a common L grid and summed before the detector trace is launched.

4.2 Refinement

The selected-rod profile is fit directly. The known Bragg positions for the convolution of polytypes in a Bragg rod define the centers of the profile terms inherited from the fixed detector geometry and ordered structural model. The disorder parameters control the redistribution of intensity among those terms and the diffuse between-peak signal through Eqs. (**??**)–(24).

Beam and detector terms, sample alignment, mosaicity, profile broadening, lattice constants, and the ordered PbI_2 rod intensities are fixed first by the prior method for ordered structures. The disorder model is then asked to explain the remaining systematic diffuse signal on selected m/L rods. This ordering is important because a diffuse feature should be interpreted as stacking information only after the detector geometry and orientation distribution have already been constrained by the ordered Bragg features.

The residual minimized in this stage has the form

$$r_j = \frac{I_j^{\mathrm{meas}} - I_j^{\mathrm{model}}(\boldsymbol{\theta}_{\mathrm{dis}})}{\sigma_j}, \tag{27}$$

where $\boldsymbol{\theta}_{\mathrm{dis}}$ contains the slip probability, finite-stack length, declared mixture weights, and any explicitly allowed nuisance scale or background terms. j indexes sampled points along the selected rod profile, and σ_j is the assigned intensity uncertainty for point j. The fit is accepted only if it improves the selected rod profile without destroying the ordered detector-space agreement established by the upstream refinement stages.

Disorder Series Fitting

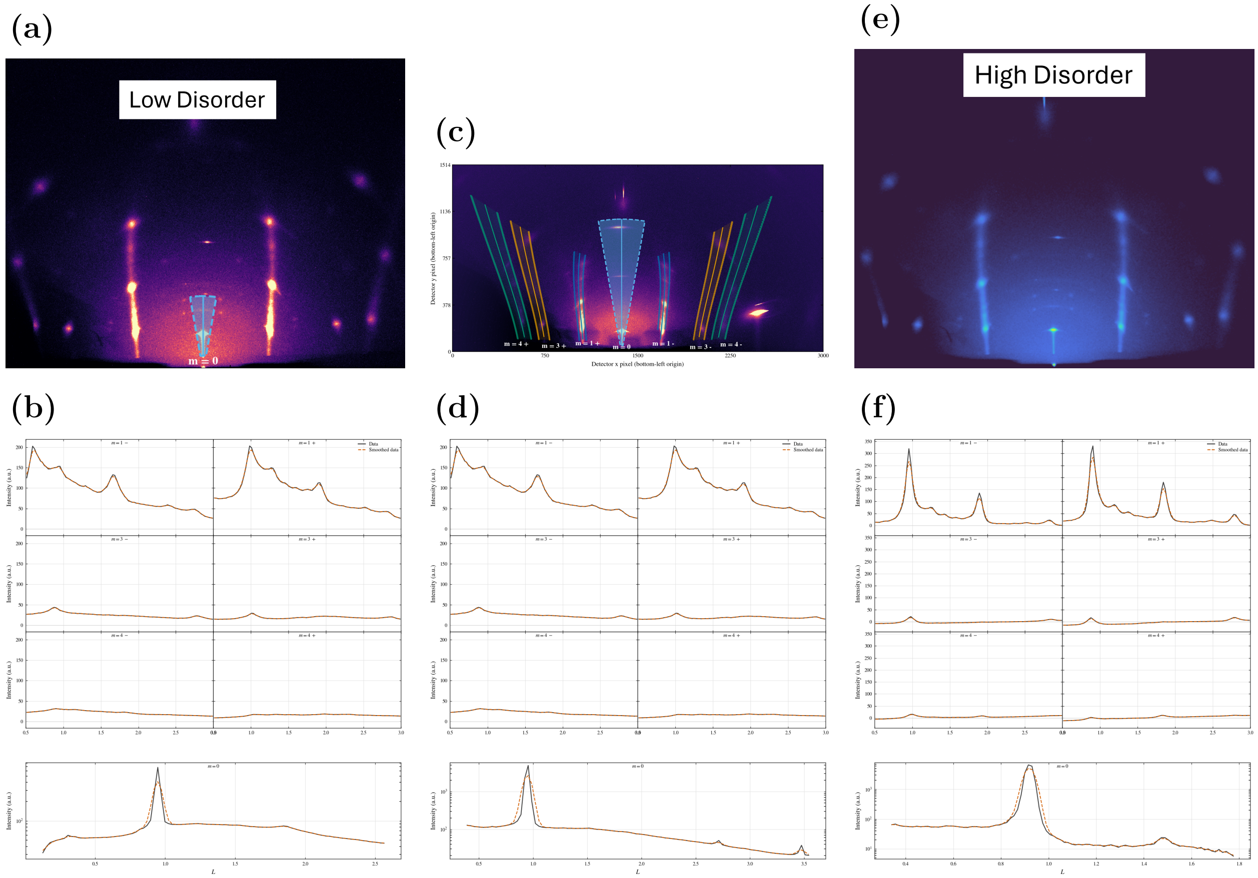

Figure 10: PbI_2 diffuse-rod validation at $\theta_i = 4°$ for samples spanning increasing stacking disorder. The left column, (a,b), shows a less-disordered sample with prominent peaks. The middle column, (c,d), shows a sample with intermediate disorder seen from the diffuse between the peaks. The right column, (e,f), shows a highly disordered sample with pronounced stacking faults. In the top row, measured detector images are shown with selected constant-m extraction regions overlaid. Solid colored curves mark the projected rod centerlines, semi-transparent regions mark the integrated detector pixels, and dashed outlines mark the integration-band boundaries. The labels m identify two-dimensional powder rod families, with signs distinguishing the two detector-side branches of the same family. In the bottom row, extracted profiles are plotted versus L after applying the same projection to measured and calculated images. Black solid curves are measured data and orange dashed curves are simulations. The progression across the three samples shows how increasing stacking disorder modifies the diffuse-rod profiles while testing the stacking-disorder model, detector geometry, mosaic broadening, and relative structure-factor weighting of the constant-m rod families.

In Fig.10 we see a series of PbI_2 samples with increasing disorder evident by the sharpness of the peaks in (a) and the increased diffuse signal between the peaks in (e). We observe a few unique signatures of these disordered powders. The m=0 rod does not express diffuse as its the specular direction in which no faults occur. The m=3 rods also do not express diffuse although due to the stacking fault leaving exceptions for when 2h+k=0. The appearance of higher order m=0/L peaks at increased disorder is not due to the stacking faults but rather an increased mosaic from growth conditions.

The ordered-baseline and disorder parameters used for the selected-rod comparison are summarized in Table 1. These values should be read as selected-rod descriptors of the present model state rather than universal PbI_2 constants. The clean ordered baseline keeps the average structure stoichiometric, while the diffuse rod intensity is assigned to stacking correlations.

Table 1: PbI_2 ordered-baseline and disorder parameters used in the selected-rod comparison. PbI_2 displacement parameters are listed as fitted sample parameters because site-resolved Pb and I ADP values are not widely or consistently reported in a directly transferable convention.

Parameter class	Fitted value	Literature comparison	Role in the selected-rod model
Lattice and coordinates	$a = 4.5583\,\text{Å}$, $c = 6.985\,\text{Å}$; Pb at $(0, 0, 0)$; I1 at $(1/3, 2/3, 0.2677)$ and I2 at $(2/3, 1/3, 0.7323)$.	Lattice constants remain within the 2H single-crystal structural spread, and the iodine coordinate follows the corrected $z \simeq 0.268$ value.[41, 42]	Fixes the ordered Bragg positions and the $F^2_{hk}(L)$ envelope before stacking disorder is introduced.
Occupancies	Weakly floating test returned $o_{\text{Pb}} = 0.999$, $o_{\text{I1}} = 1.001$, and $o_{\text{I2}} = 1.000$; the ordered baseline was then evaluated with full occupancies.	The clean ordered model uses full occupancies; the density-deficient occupancy reported for a Palosz-type sample is sample-specific and was not used as the default model.[42, 43]	Keeps the PbI_2 average structure stoichiometric while the diffuse rod intensity is assigned to stacking disorder.
Directional ADPs	$U_R/U_z = 0.0105/0.0185$ for Pb and $0.016/0.024\,\text{Å}^2$ for I1 and I2.	Site-resolved Pb and I ADPs are not widely or consistently reported for direct transfer; the qualitative inequality $U_z(\text{Pb}) > U_R(\text{Pb})$ is retained.[41, 44]	Moderates relative ordered intensities and is interpreted as a sample-specific displacement refinement rather than a universal PbI_2 displacement constant.
Stacking disorder	$p = 0.18$, $N = 32$ layers, and a 2H:6H motif fraction of $0.76 : 0.24$.	Treated as effective selected-rod descriptors of the fitted model class rather than universal PbI_2 structural constants.	Converts the ordered rod envelope into broadened between-peak intensity along the selected Q_z rod.

The physical interpretation is therefore deliberately limited. The fitted slip probability, finite-stack length, and 2H:6H mixture describe the stacking correlations required by the measured selected-rod intensity under the stated Hendricks–Teller convention.

5 Refinement workflow

The workflow is placed after the ordered and PbI_2 result sections because the reader has now seen what the parameters are meant to explain. The refinement is staged rather than fully global from the beginning. Many parameters affect similar detector-space observables: detector distance shifts peak positions, sample alignment changes the apparent incident condition, mosaicity changes profile widths and off-condition intensity, and structure factors change relative peak heights. A simultaneous fit can therefore hide physical errors by compensating one parameter group with another. The staged workflow fixes the most instrumental quantities first, then introduces sample orientation, mosaicity, ordered intensity, and finally stacking disorder where needed.

The first distinction is between instrumental parameters and sample parameters. Instrumental parameters describe the beam and detector and should be shared across samples collected under the same experimental configuration. Sample parameters describe the film: incidence condition, sample offset, mosaic distribution, structural intensity, and, for PbI_2, stacking disorder. The refinement sequence in Table 2 uses the detector-space observable most sensitive to each group before that group is used in the final image projection.

Table 2: Staged refinement workflow. Steps 1–5 are used for the ordered Bi_2Se_3 and Bi_2Te_3 validation. Step 6 is the added PbI_2 extension for stacking-fault diffuse scattering. All sample images are dark-subtracted before the listed observables are formed, and the final measured/calculated comparison applies the same detector-space projection to data and simulation.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry $(\theta_i, \delta, z_B, z_S)$, in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, specular-tail checks, and local profile shapes	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes, off-specular arcs, and specular tails
5	Ordered-film validation	Selected detector-space Bragg regions and Q_z projections from fixed-m trajectories	Image scale factors and ordered structure-factor parameters	Relative Bragg intensities and measured/calculated line-shape agreement
6	PbI_2 diffuse extension	Fixed-Q_R diffuse regions, ordered-baseline residuals, and projected Q_z profiles	Stacking-fault probabilities or layer-pair correlation parameters	Diffuse intensity along Q_z and separation of ordered and disorder-derived scattering

The instrumental stages fix where an ideal reflection should appear before any sample-specific physics is refined. Direct-beam images define the incident phase space, including beam width and angular divergence. A powder calibrant fixes the detector mapping: distance, beam center, and detector tilts. These two stages are intentionally separated from sample fitting because an incorrect detector geometry can otherwise be absorbed into apparent sample tilt, mosaic broadening, or shifted reciprocal-space trajectories.

The alignment stage uses indexed sample reflections to determine how the film sits in the beam. This stage fixes the nominal incident angle, sample offsets, in-plane rotation, and goniometer misalignment well enough that calculated Bragg-rod trajectories can be overlaid on the detector. Only after this step does it make sense to refine the mosaic distribution, because mosaicity should explain profile shape and off-condition intensity rather than compensate for a misplaced trajectory.

The mosaicity stage constrains the shared Gaussian-plus-Lorentzian angular distribution. Local off-specular profiles constrain the narrow core, while the specular-family tails and low-L features test the long-tail component. The ordered-film validation then applies the same projection to measured and calculated images and compares profiles along fixed-m trajectories. At that point the structure-factor parameters and image scale factors are constrained by relative intensities, not by detector placement alone.

For PbI_2, the ordered workflow is not discarded. It provides the geometry, projection, and mosaic baseline needed before diffuse scattering can be interpreted. The additional stacking-disorder stage asks whether a physically defined layer-correlation model accounts for the remaining diffuse intensity along fixed-Q_R cylinders. This keeps the PbI_2 result tied to the same detector-space framework while making clear which extra parameters belong specifically to stacking faults.

Implementation details, including lookup tables, Monte Carlo sampling, sub-pixel remapping, parameter-correlation plots, and the propagated uncertainty from 2θ–ϕ and Q transformations, should be documented in the supporting information. The main-text role of the workflow is to make the logic of the refinement reproducible: determine instrumental geometry first, determine sample orientation second, refine mosaic/profile effects third, and only then interpret ordered or diffuse intensity differences as structural information.

6 Discussion and conclusion

The ordered Bi_2Se_3 and Bi_2Te_3 films validate the mosaic model and detector-space forward model. A single calculation accounts for detector geometry, finite beam phase space, sample alignment, mosaicity, resolution, and ordered structure-factor weighting, then compares measured and calculated profiles along the same fixed-m trajectories. The key result is direct line-shape agreement in detector-derived projections, including features that would be difficult to interpret from a one-dimensional reduction alone.

Keeping the comparison in detector space is essential because the constraining intensity is distributed among off-specular arcs, cap-like $m = 0$ features, weak off-condition specular peaks, and bright low-L specular-region intensity that a conventional one-dimensional reduction would obscure. Because the calculated image includes the same effects and is projected in the same way as the measurement, the overlay tests the complete experimental model. The off-condition $m = 0$ peaks show why a narrow Gaussian-like mosaic core is insufficient and why the Lorentzian-like tail is required, while the near-origin low-L feature emphasizes the importance of finite beam and wavelength phase space.

The PbI_2 section outlines how the same framework can be extended from ordered Bragg line-shape validation to diffuse scattering from stacking disorder. In that extension, the detector geometry, beam phase space, mosaicity, and projection procedure remain the same, but the reciprocal-space intensity is generated from a stacking-fault correlation model rather than from a perfectly periodic ordered stack. This separation is important: diffuse intensity should be interpreted as stacking information only after geometry and mosaicity have been constrained by the ordered-film workflow.

The practical refinement workflow is therefore part of the scientific argument. Instrumental parameters are determined first, sample alignment second, mosaic and profile parameters third, and ordered or diffuse intensity parameters last. This ordering reduces parameter compensation and makes the final data/model comparison interpretable. The resulting manuscript structure follows the same logic: show the detector features that require each model ingredient, explain the physical effect in reciprocal-space terms, and then use measured/calculated overlays as the validation.

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